KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 9 STANDARD

EXTENSION BOOKLET

FOR FAST FINISHERS · CHALLENGE QUESTIONS

CHAPTER 10

YEAR 9 MATHEMATICS · EXTENSION

Surface area and volume

One page of harder questions for each lesson from 10.6 on. Finish the lesson in your Chapter 10 booklet first, then come here.

LESSON	EXTENSION TOPIC	PAGE	DONE
10.6	Circumference and arc length	2	☐
10.7	Areas of circles and sectors	3	☐
10.8	Surface area of a prism	4	☐
10.9	Surface area of a cylinder	5	☐
10.10	Volumes of prisms and cylinders	6	☐

EVERY CARD HAS A TYPE
Understanding · Reasoning · Problem solving

MIXED IN
Review 10.4 questions from earlier lessons

NO DIAGRAM?
Draw a diagram first, in the dotted space

CHALLENGE
★☆☆ to ★★★

NAME _____ CLASS _____ TEACHER _____

Artwork: the Chapter 10 spiral from Seahorse Valley, part of the Mandelbrot set, coloured silver.

Circumference and arc length

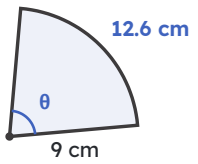
FAST FINISHERS

EXTENSION Finished lesson 10.6? Try these. Show your working in each box. Answers with π to **one decimal place**.

Understanding | Reasoning | Problem solving

1 Understanding ★★☆

This sector has a radius of 9 cm and an arc length of 12.6 cm.



a Find the circumference of the whole circle.

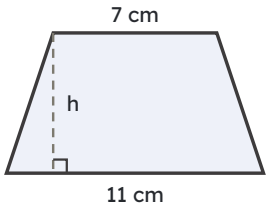
b Find the angle θ, to the nearest degree.

c Find the perimeter of the sector.

a _____ b _____ c _____

2 Understanding | Review 10.5 ★☆☆

This trapezium has an area of 54 cm². Find its height h.



h = _____

3 Problem solving | Draw a diagram ★★★

A running track has two straight sides, each 100 m long, joined by two semicircular ends. One lap of the inside edge is 400 m.

a Find the diameter of the semicircular ends.

b The outside lane has the same straights, but its ends have a diameter 2.4 m longer. How much further is one lap of the outside lane?

a _____ b _____

4 Problem solving ★★☆

The minute hand of a clock is 15 cm long.

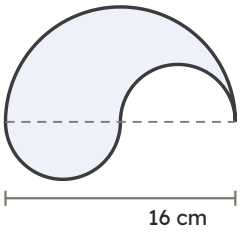
a How far does the tip of the minute hand travel in 20 minutes?

b How far does the tip travel in one whole day? Give your answer in metres.

a _____ b _____

5 Reasoning ★★★

This shape is made from a semicircle with diameter 16 cm and two semicircles with diameter 8 cm.



a Find the perimeter of the shape.

b Show that it is the same as the circumference of a circle with diameter 16 cm.

a _____

6 Reasoning | Draw a diagram ★★★

A rope fits tightly around a basketball with a diameter of 24 cm. The rope is made 1 m longer and held in a circle, the same distance from the ball all the way round.

a How big is the gap between the rope and the ball?

b Would the gap be bigger, smaller or the same for a rope 1 m longer than the equator? Explain.

a _____

b _____

Areas of circles and sectors

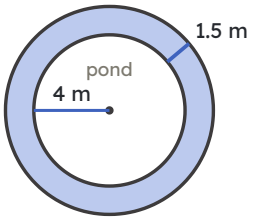
FAST FINISHERS

EXTENSION Finished lesson 10.7? Try these. Show your working in each box. Answers with π to **one decimal place**.

Understanding | Reasoning | Problem solving

1 Reasoning ★★☆

A circular pond with a radius of 4 m has a path 1.5 m wide around it.



a Find the area of the path.

b Lily says the path's area is $\pi \times 1.5^2$. Explain why she is wrong.

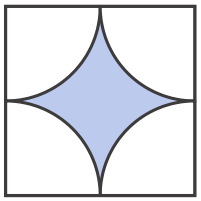
a _____

b _____

_____

2 Understanding ★★☆

A square tile of side 10 cm has a quarter circle of radius 5 cm at each corner.



a Find the area of the shaded part.

b What percentage of the tile is shaded?

a _____ b _____

3 Reasoning ★★☆

A large pizza has a diameter of 30 cm and costs $18. A small pizza has a diameter of 20 cm and costs $9.

a Find the area of each pizza.

b Which pizza is better value for money? Show your reasoning.

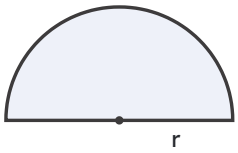
a _____

b _____

_____

4 Understanding Review 10.6 ★★★

A semicircle has a perimeter of 36 cm.
(Perimeter = curved part + diameter.)



a Find its radius r.

b Find its area.

a _____ b _____

5 Problem solving Review 10.4 Draw a diagram ★★☆

A rectangular lawn 12 m long and 9 m wide has a path 1 m wide all the way around the outside.

a Find the area of the path.

b Paving costs $55 per square metre. Find the cost of paving the path.

a _____ b _____

6 Problem solving Draw a diagram ★★★

A goat is tied to a post by a rope 5 m long.

a The post is halfway along a long straight fence. What area of grass can the goat reach?

b The post is moved to the outside corner of a large shed, so the goat can walk three-quarters of the way around the post. Find the new area.

a _____ b _____

Surface area of a prism

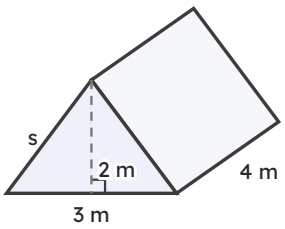
FAST FINISHERS

EXTENSION Finished lesson 10.8? Try these. Show your working in each box. Answers with π to **one decimal place**.

Understanding | Reasoning | Problem solving

1 Problem solving ★★☆

A tent is a triangular prism with a floor. Its ends are isosceles triangles.



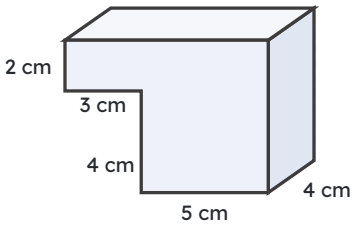
a Use Pythagoras' theorem to find the slant length s.

b Find the area of canvas needed to make the tent, floor included.

a _____ b _____

2 Understanding ★★★

Some lengths of this prism are not marked. Work them out first.



a Find the area of the front face.

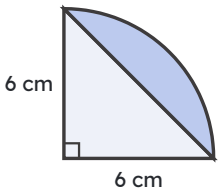
b Find the perimeter of the front face.

c Find the surface area of the prism.

a _____ b _____ c _____

3 Understanding Review 10.7 ★★☆

A straight cut across a quarter circle of radius 6 cm makes the shaded segment. Find the area of the segment.



Area = _____

4 Reasoning ★★☆

Box A measures 2 cm × 3 cm × 4 cm. Box B measures 1 cm × 4 cm × 6 cm.

a Show that the two boxes have the same volume.

b Find the surface area of each box.

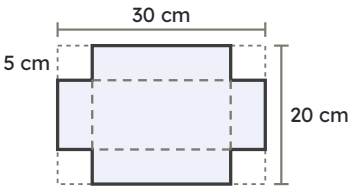
c Which box needs less cardboard? Why might a company choose it?

a _____ b _____

c _____

5 Reasoning ★★☆

An open box (no lid) is made from a card 30 cm by 20 cm: a 5 cm square is cut from each corner and the sides are folded up.



a Find the length, width and height of the box.

b Find the outside surface area of the box.

c Explain a quick way to get the answer to b from the card.

a _____ b _____

c _____

6 Problem solving Draw a diagram ★★★

A room is 5 m long, 4 m wide and 2.7 m high. Its four walls and the ceiling are painted. There is a door 2 m by 0.9 m and a window 1.5 m by 1.2 m.

a Find the area to be painted.

b 1 L of paint covers 12 m². How many 4 L tins are needed for two coats?

a _____ b _____

Surface area of a cylinder

FAST FINISHERS

EXTENSION Finished lesson 10.9? Try these. Show your working in each box. Answers with π to **one decimal place**.

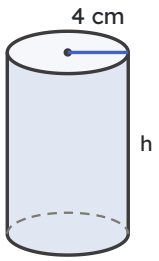
Understanding | Reasoning | Problem solving

1 Understanding ★★☆

The curved surface area of this cylinder is 301.6 cm².

a Find its height h.

b Find its total surface area (top and bottom included).



a _____ b _____

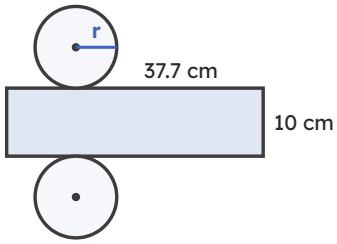
2 Reasoning ★★☆

This is the net of a cylinder.

a Explain why the rectangle is 37.7 cm long.

b Find the radius r of each circle.

c Find the total surface area.



a _____

b _____ c _____

3 Reasoning ★★☆

Cylinder P has radius 5 cm and height 10 cm. Cylinder Q has radius 10 cm and height 5 cm.

a Show that their curved surface areas are equal.

b Which cylinder has the greater total surface area, and by how much?

c Explain why, using the parts of each net.

a _____ b _____

c _____

4 Problem solving Draw a diagram ★★★

A water trough is half a cylinder, open at the top. Its radius is 40 cm and it is 2 m long.

a Find the area of its curved part, in m².

b The two ends are half circles. Find the total area of metal needed, in m².

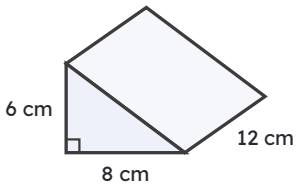
a _____ b _____

5 Understanding Review 10.8 ★★☆

The ends of this prism are right-angled triangles.

a Use Pythagoras' theorem to find the missing side of the triangle.

b Find the surface area of the prism.



a _____ b _____

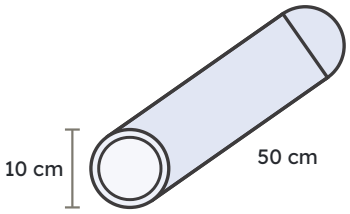
6 Problem solving ★★★

A pipe is 50 cm long. Its outside diameter is 10 cm and its inside diameter is 8 cm.

a Find the area of the outside curved surface.

b Find the area of the inside curved surface.

c Find the total surface area, including the two ring-shaped ends.



a _____ b _____ c _____

Volumes of prisms and cylinders

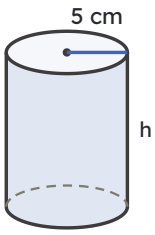
FAST FINISHERS

EXTENSION Finished lesson 10.10? Try these. Show your working in each box. Answers with π to **one decimal place**.

Understanding | Reasoning | Problem solving

1 Understanding ★★☆

This cylinder holds exactly 1 L of water when it is full. (1 L = 1000 cm³)



a Find its height h.

b How tall would a 2 L cylinder with the same radius be?

a _____ b _____

2 Reasoning ★★☆

A cylinder has radius 4 cm and height 10 cm.

a Find its volume.

b Find the volume when the radius is doubled and the height is halved.

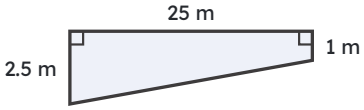
c What happens to the volume? Explain why.

a _____ b _____

c _____

3 Problem solving ★★★

A swimming pool is 25 m long and 10 m wide. The diagram shows its side: it is 2.5 m deep at one end and 1 m deep at the other.



Side view, not to scale

a Find the area of the side of the pool (a trapezium).

b Find the volume of the pool in litres. (1 m³ = 1000 L)

c A pump fills the pool at 250 L per minute. How long does it take, in hours and minutes?

a _____ b _____ c _____

4 Reasoning Review 10.9 ★★☆

A metal cube with edges of 6 cm is melted down and made into a cylinder with a radius of 3 cm.

a Find the height of the cylinder.

b Do the cube and the cylinder have the same surface area? Show your working.

a _____ b _____

5 Problem solving Review 10.1 Draw a diagram ★★★

15 mm of rain falls on a flat roof 12 m long and 8 m wide. All of it runs into a cylindrical tank 2 m across.

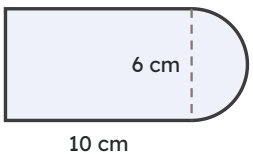
a How many litres of water fall on the roof?

b How many centimetres does the water in the tank rise?

a _____ b _____

6 Understanding Review 10.7 ★★☆

This shape is a rectangle with a semicircle on one end.



a Find its area.

b The shape is the cross-section of a prism 20 cm long. Find the volume of the prism.

a _____ b _____